



FAST National University
of Computer and Emerging Sciences
Operating Systems (CS2006)

Quiz-1: (CLO - 1)

Max. Marks: 02

Name: _____

Student ID: _____

Instructions:

Max Time: 20 Minutes

- Don't write with Pencil, Even Diagrams should be made with Pen otherwise marks will be deducted
 - Answer Question in the context of Linux
 - Instructions in Questions should be followed strictly
 - Write your answer after each question -
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Question 1: (Maximum Single Line Answers, if exceeds marks will be deducted) (Marks: 01 = 0.25*4)

I. Explain the purpose of the O_CREAT flag in open ().

Answer: The O_CREAT flag tells open() to create the file if it does not already exist.

II. A shell program needs to run the command ls -l. Which system call(s) would it use?

Answer: fork(), then exec() (like execvp()), and often wait().

III. A web server handles multiple clients by creating a new process per connection. Which system call creates the new process?

Answer: fork() creates the new process for each client connection.

IV. What security risk exists if applications could access hardware directly?

Answer: Direct hardware access would let malicious programs bypass security controls, compromising system stability, stealing data, or hijacking resources without authorization.

Question 2: (Code Completion and Diagram)

(Marks: 01 = 0.25 + 0.25 + 0.5)

I. What will be the output of this program?

```
pid_t pid = fork();
if (pid == 0) printf("Child");
else printf("Parent");
```

Parent		Child
Child	OR	Parent

II. Complete the code to write "Hello World" to a file:

```
int fd = open("output.txt", O_WRONLY | O_CREAT, 0644);
write(fd, _____, _____);
close(fd);
```

Answer: 'Hello World', strlen('Hello World') or 11

III. Draw and label the sequence of events when a user program calls **read(fd, buffer, size)**:

